Planning-Guided Failure-Triggered Recovery via Imitation Learning for Dynamic Social Navigation

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Abstract—Classical navigation stacks are reliable in routine mapped environments but can fail in reciprocal interactions with moving people. End-to-end learned navigation can address such interactions, yet replacing a mature planner everywhere changes behavior even where no recovery is needed. We study a bounded alternative: an observable failure detector invokes a learned policy only during collision risk, freezing, oscillation, or deadlock. The policy chooses among 21 temporary subgoals and four interpretable modes, while a planning-derived mask rejects invalid actions and Nav2 executes each selected goal. A privileged short-horizon expert supplies imitation labels and DAgger covers learner-induced recovery states. During validation we found that simulator command acceptance was insufficient evidence that sensed geometry, localization, and evaluation truth agreed. The corrected fallback reads robot and pedestrian evaluation poses from Gazebo, separates rendered pedestrians from contact dynamics, and uses pose-derived localization. Across three fixed high-density crossing-flow seeds, DWB reached two goals and collided once; triggered DAgger physically reached all three goals and increased median minimum human-centre distance from 0.927 to 1.027 m, but increased median terminal time from 91.7 to 128.4 s. Two separately versioned high-density checks also converted DWB collisions in temporary- and group-blocking scenarios into physical goal reaches. These executions establish preliminary cross-family feasibility and a safety–time trade-off, not statistical superiority; a locked multi-scenario evaluation remains required.

I. Introduction

Classical navigation offers explicit maps, reusable planners, and predictable nominal behavior, but local optimization can freeze or oscillate in interactive bottlenecks. Learned policies can encode interaction patterns, but continuous end-to-end replacement expands the learned component's authority and makes failures harder to interpret.

We instead confine learning to the failure boundary. Figure 1 shows the hierarchy. A classical planner owns nominal PointGoal navigation. An observable detector and hysteretic state machine trigger a bounded recovery option. A compact policy chooses an interpretable temporary target or mode from a fixed action set, subject to one mask shared by data generation and deployment. Once progress resumes, the original task goal is reissued.

The current contribution is threefold: (i) a failure-triggered architecture that preserves classical nominal control; (ii) an automated privileged planning expert that labels masked temporary-goal choices; and (iii) an imitation pipeline using behavior cloning and DAgger. We do not claim a first recovery system or a formal safety guarantee. This evidence-linked draft reports the corrected simulator-validity result and does not reuse pre-verified-pose comparisons as performance evidence.

II. Related Work

Arena-Bench and Arena-Rosnav provide reproducible interfaces for comparing model-based and learned navigation in dynamic environments [1], [2], [3], [4]. All-in-One learns to select among navigation planners [5]; DR-MPC learns a residual around model-predictive control [6]. Both motivate hybrid navigation but differ from failure-only intervention.

Learned navigation recovery predates this work. Del Duchetto et al. learn failure detection and local recovery from human demonstrations [7]. Mohammad et al. proactively predict planning failure and search for a recovery state [8]. Recent failure-aware learning uses unsafe experience to shape value estimation while restricting policy learning to successful demonstrations [9]. Our narrower distinction is automatic recovery demonstrations from privileged rollout planning, temporary-subgoal decisions, and aggregation on learner-induced recovery states. DAgger is the standard mechanism for mitigating sequential distribution shift [10].

III. Method

A. Failure-Triggered Hierarchy

Let g be the original goal and let the classical stack produce command u_b. From observable history o_t, the detector returns scores for collision risk, freeze, oscillation, and deadlock. Recovery begins after the maximum score exceeds τ_{on} and exits only after it remains below $\tau_{\text{off}} < \tau_{\text{on}}$ while valid goal progress resumes. Cooldown, option duration, and recovery-count limits prevent rapid switching.

B. Action Space and Mask

The action space contains temporary goals at radii $\{0.6, 1.0, 1.4\}$ m and bearings $\{-90, -60, -30, 0, 30, 60, 90\}^\circ$, plus WAIT, BACKUP, REPLAN, and CONTINUE. The mask rejects occupied, disconnected, LiDAR-occluded, rear-unobserved, or path-divergent motion. During latched collision risk it also blocks unsafe continuation and applies endpoint and swept-segment clearance. Masking precedes softmax, so invalid actions cannot be selected while a valid fallback remains.

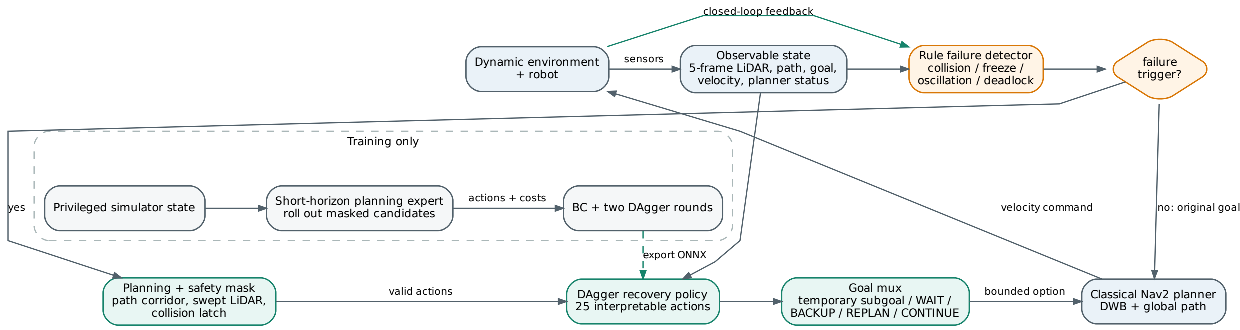

Fig. 1. Closed-loop architecture. Privileged state and expert rollouts are used only during training. At test time, the policy receives observable state and may only choose actions admitted by the planning and safety mask.

An independent stopping layer uses

$$d_{\text{stop}} = \frac{v^2}{2a_{\text{brake}}} + v\ell + m, \tag{1}$$

where ℓ is control latency and m an empirical margin. It may override learning but is not a formal collision-avoidance certificate.

C. Privileged Expert and Imitation

During training only, the expert receives the static grid and true pedestrian states. Each legal candidate is rolled out with differential-drive kinematics and predicted human motion. It minimizes collision, progress, social distance, path rejoin, length, smoothness, time, and action-switch costs; collision and masked actions receive hard penalties. The policy uses a one-dimensional CNN over five 180-beam scans and an MLP over observable navigation state. Behavior cloning minimizes expert cross entropy, and DAgger relabels policy-visited training states with the privileged expert.

IV. Experimental Setup

Experiments use ROS2 Humble, Nav2 DWB, and a pinned Arena Humble Gazebo profile. The installed profile lacks a complete HuNav dependency chain, so deterministic LiDAR-visible fallback pedestrians are disclosed. The robot receives a known static map and a planar scan downsampled to 180 beams. Train, validation, and test splits are scenario-separated.

Fallback pedestrians are non-static, gravity-disabled rendered cylinders without contact response; overlap is classified geometrically from Gazebo model poses. This prevents teleported proxy bodies from imparting non-human impulses to the Jackal while preserving GPU-LiDAR visibility. Gazebo's pose-derived odometry supplies the declared known-pose localization interface; a real mapped deployment would replace it with localization such as AMCL. An evaluation-only gate projects human

TABLE I
Verified-pose medium crossing-flow validation pair. Goal distance and human clearance use Gazebo model feedback; this is single-seed execution evidence only.

Method	Outcome	Time [s]	d_g^{phys} [m]	$d_{\min}$ [m]	Recovery
Classical DWB	COLLISION	29.5	14.074	0.665	0
Triggered DAgger	GOAL REACHED	98.4	0.299	1.203	117

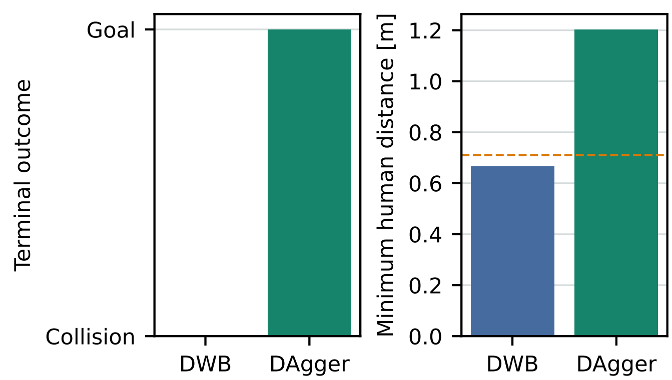

Fig. 2. First verified-pose pair. DWB collides; learned recovery reaches the physical goal while maintaining greater human clearance.

centres into the scan and checks that near-human surfaces are visible at expected bearings. Human privileged data are never policy inputs. All valid outcomes are retained; simulator/reset failures are reported separately. The reported pilot pairs use validation scenarios and are not a final test evaluation.

V. Corrected Validation Check

The static-proxy implementation produced an impossible observation: privileged evaluation placed a pedestrian centre 0.697 m from the robot while the LiDAR sector at that bearing remained 1.473 m away. A non-static kinematic-link revision still allowed model pose requests to diverge from rendered geometry. A later dynamic

TABLE II
High-density crossing-flow validation pilot on three fixed seeds. Values are descriptive medians; $n = 3$ is insufficient for significance testing.

Method	Goal	Collision	Time [s]	$d_{\min}$ [m]	Recovery
Classical DWB	2/3	1/3	91.7	0.927	0.0
Triggered DAgger	3/3	0/3	128.4	1.027	308.0

TABLE III
Cross-family high-density validation examples (DWB/full hierarchy). Each row is one internally same-commit pair; values are descriptive and are not pooled for inference.

Scenario	DWB	Hierarchy	Time [s]	$d_{\min}$ [m]	Recovery
Temporary blockage	COLLISION	GOAL REACHED	38.0/80.9	0.684/0.781	132
Group blocking	COLLISION	GOAL REACHED	46.0/128.9	0.689/1.019	395
Overtaking	COLLISION	GOAL REACHED	82.0/178.4	0.685/1.156	1133
Blind corner	COLLISION	TIMEOUT	49.5/180.0	2.710/1.876	1088

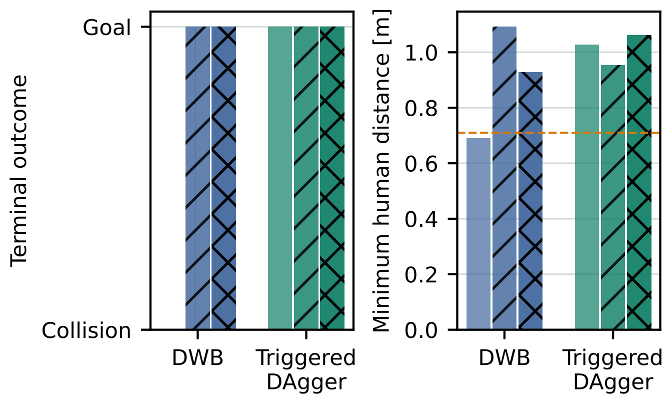

Fig. 3. Three-seed high-density pilot under one frozen code revision; bars within each method are seeds 2201, 2202, and 2220 from left to right. Triggered DAgger converts one DWB collision and preserves two goal reaches. The dashed line is the 0.71 m geometric robot–human collision threshold.

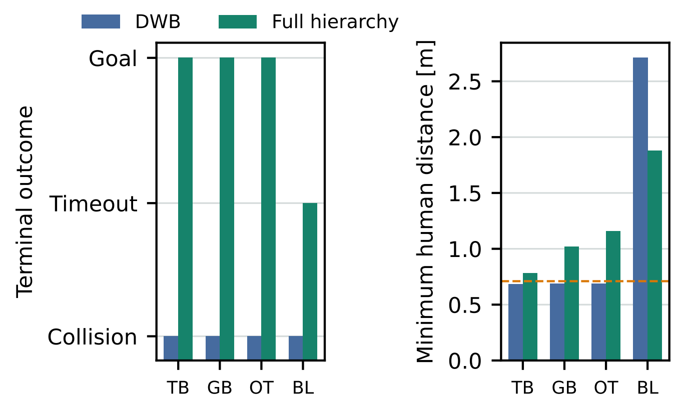

Fig. 4. Four cross-family validation examples: temporary blockage (TB), group blocking (GB), overtaking (OT), and blind corner (BL). Each pair uses one scenario, seed, and code revision for both methods. The dashed line marks the 0.71 m geometric collision threshold. These single pairs are not pooled for significance testing.

collision-body revision pushed the Jackal, and wheel-integrated odometry declared arrival while the physical robot remained 1.06 m from the goal. Consequently, every pre-verified-pose comparison is retained only for debugging. The final path reads evaluation poses from Gazebo's dynamic-pose topic, uses rendered contactless pedestrians, and checks physical goal distance independently of localized distance.

Table I and Fig. 2 summarize the first verified-pose pair from commit 2164e08. DWB collided at 29.47 s with 0.665 m minimum centre distance. The learned hierarchy physically reached the goal at 98.43 s after 117 non-CONTINUE recovery samples, with 1.203 m minimum centre distance and 0.299 m final physical goal error. Its sensor gate passed 30/30 near-human frames; Base passed 26/26. Maximum localization–physical position disagreement was 0.128 m.

The current high-density pilot in Table II and Fig. 3 uses three independent scenario seeds under commit 8577ff1. On seed 2201, DWB collided at 29.50 s and triggered DAgger physically reached the goal at 128.40 s with 0.215 m terminal physical error. On seeds 2202 and 2220, both methods reached the goal; DAgger required 149.92 and 118.41 s versus DWB's 95.20 and 91.67 s. Median minimum centre distance changed from 0.927 to 1.027 m. Learned runs used 308 median non-CONTINUE samples and included WAIT, BACKUP, and temporary subgoals. All six sensor-consistency gates passed; the least-visible run still passed 77/80 near-human frames. Thus recovery converts one observed collision and preserves two nominal successes, but is consistently slower. With $n = 3$, the one discordant outcome is not statistically significant and no general performance claim is justified.

Table III and Fig. 4 add four high-density, single-seed checks from distinct scenario families. In temporary blockage, DWB collided at 38.00 s and the full hierarchy reached the physical goal at 80.92 s with 0.195 m terminal error and 0.781 m minimum human distance. The escape in this historical trace was dominated by deterministic safety turn/forward commands, so it does not isolate the learned policy. In the current same-commit group-blocking pair, DWB collided at 46.02 s while the hierarchy reached the physical goal at 128.94 s with 0.211 m terminal error and 1.019 m minimum human distance. The learned policy selected temporary-subgoal actions 3 and 11 alongside separately logged STOP, BACKUP, TURN, and FORWARD safety modes. In the frozen overtaking pair, DWB collided at 82.02 s while the hierarchy physically reached the goal at 178.39 s with 0.224 m terminal error and 1.156 m minimum human distance. Both overtaking sensor gates passed every near-human frame, but completion had only 1.61 s timeout margin and required 1,133 recovery samples. Finally, in blind corner, DWB intersected declared static geometry after 49.48 s. The hierarchy remained collision-free and made 13.47 m progress, but timed out 8.16 m from the physical goal. This is collision avoidance, not recovery success.

Table IV adds Standard and Heuristic recovery under

TABLE IV
Same-commit group-blocking mechanism check on one validation seed. Results are descriptive; the comparison is not a statistical ablation.

Method	Outcome	Time [s]	$d_{\min}$ [m]	Recovery
Classical DWB	COLLISION	46.0	0.689	0
Standard recovery	COLLISION	45.0	0.695	0
Heuristic hierarchy	GOAL REACHED	152.4	0.841	606
Triggered DAgger	GOAL REACHED	128.9	1.019	395

TABLE V
Retained opposite-stream train failures. Revisions are diagnostic and not paired.

Variant	Progress [m]	Final 60 s [m]	Safety [%]	Backup	Subgoal
Iteration 4	2.38	0.04	64.7	146	112
Turn memory	-0.15	-0.24	67.8	118	127
Iteration 5	0.20	0.20	58.8	317	221

the identical group-blocking revision. Standard recovery collided at 44.99 s. Heuristic recovery reached the goal at 152.38 s with 0.841 m minimum human distance and 606 recovery samples; triggered DAgger reached it 23.44 s sooner with 1.019 m minimum human distance and 395 recovery samples. All four sensor gates passed at 100%. This single seed shows a concrete efficiency and clearance difference from the heuristic, but is not a statistical ablation or a generalization claim.

VI. Failure Analysis

The held-out high-density opposite-stream case exposes a boundary of the hierarchy: DWB collided after 3.50 s, whereas the selected recovery hierarchy preserved 0.838 m minimum human distance but timed out after 180 s with only 0.200 m net progress. The final minute had negative progress. We did not add this validation trajectory to training. Instead, we collected a disjoint train-split policy trajectory and tested two corrections there.

Table V and Fig. 5 show that neither intervention solved recurrent flow. A five-second cross-hazard turn commitment reduced turn-side switches from nine to zero but changed net progress from $+2.378$ m to -0.146 m. Sequence-aware DAgger with immediate freeze/deadlock WAIT exhaustion reduced action-21 occupancy and increased BACKUP/subgoal execution, yet timed out with only 0.196 m progress. Both candidates were rejected. This separates action activity from recovery: more interventions, fewer WAIT samples, or fewer switches do not imply global task progress.

VII. Limitations

The method assumes a known two-dimensional map and planar LiDAR. Its privileged expert depends on simulator state and constant-velocity pedestrian prediction. The finite discrete action set cannot express arbitrary maneuvers. The Gazebo fallback differs from real crowds and exhibits scheduling artifacts. Pose-derived simulation localization is more accurate than a real localization stack. There is no formal safety guarantee, hardware

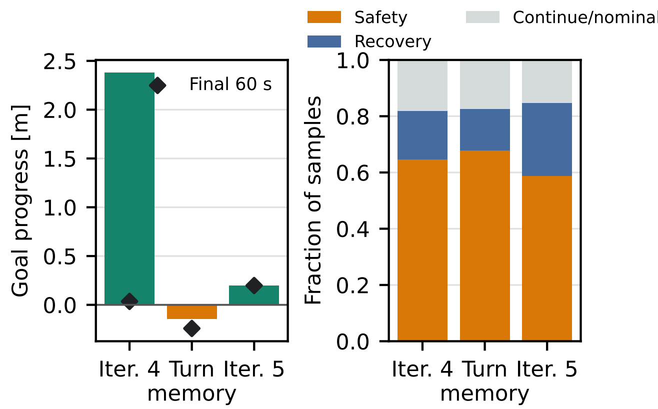

Fig. 5. Retained opposite-stream train failures across separately versioned diagnostic revisions. Bars show total goal progress and mutually exclusive control-state occupancy; diamonds show progress during the final 60 s. Turn commitment removed switches but produced negative progress, while immediate WAIT exhaustion increased active recovery without solving the timeout.

study, completed learned detector, or PPO refinement. The learned policy is conservative and substantially increases navigation time in the reported high-density pilot. Recurrent opposite-direction streams and the blind-corner case can still produce safe stagnation, and the rejected train variants show that simple anti-WAIT or anti-switch rules are insufficient. The cross-family checks contain one seed per family and use separately audited revisions, so they cannot establish generalization or statistical benefit. Corrected multi-family training data and the final locked multi-scenario test remain incomplete.

VIII. Conclusion

Failure-triggered temporary-goal recovery can be integrated with a classical ROS2 navigation stack and trained from an automated privileged planner without granting the learned policy continuous velocity authority. The simulator-validity gate exposed and corrected hidden proxy, yielding, odometry, and terminal-evidence errors. Under verified-pose implementations, the learned hierarchy converted one DWB collision and preserved two DWB goal reaches across three fixed high-density crossing-flow seeds, and separately recovered one temporary-blockage, one group-blocking, and one overtaking DWB collision. The group-blocking and overtaking runs include actual learned temporary-subgoal selections. All recovered executions took longer, and blind corner remained a timeout despite collision avoidance. This is mixed cross-family execution evidence rather than a general claim. The required next steps are broader corrected-data training and a locked paired multi-scenario evaluation.

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